

SEMESTER-III

UES034: MEASUREMENT SCIENCE AND TECHNIQUES				
	L	T	P	Cr
	3	1	2	4.5
Course Objective: To introduce the basic concepts of instruments, errors, system characteristics, sensors, transducers, and signal conditioning.				
Syllabus Measuring Instruments: Functional elements of an Instrument, Active and Passive transducers, Analog and digital modes of operation, Null and Deflection Methods, Input-Output Configurations of Measuring Instruments and Instrument systems. Error Analysis: Types of errors, Uncertainty analysis, Statistical analysis, Gaussian error distribution, Chi- Square test, Correlation coefficient, Student's t-test, Method of least square, Curve fitting by the method of least squares- fitting of straight lines, second degree parabolas and more general curves. Static Characteristics: Calibration, Accuracy, Precision. Combination of errors, static sensitivity, Linearity, Threshold, Resolution, Hysteresis and dead space. Span. Dynamic Characteristics: mathematical model of measurement system, Digital simulation methods for dynamic response analysis. Transfer Function, Zero order Instrument, First order Instrument, Step response, ramp response, frequency response, Impulse response of first and second order instruments. Dead Time elements, Logarithmic plotting of frequency- response curves, Response of general form of instrument to a periodic and transient input. Sensors/Transducers: Basic principle and applications of Resistive, Inductive, Capacitive, Piezoelectric and their Dynamic performance, Hall-Effect, Photo emissive, Photo Diode/ Photo Transistor, Photovoltaic, LVDT, Strain Gauge. Introduction of Signal Conditioning: Amplification, filtering, converting, range matching, isolation, linearization, attenuator, surge protection, excitation, AC coupling.				
Laboratory Work: Experiments based on first order, second order and time parameters, measurement of signal, sensors and transducers.				
Course Learning Objectives (CLO) The students will be able to: 1. Explain the concepts of measurement and its analysis. 2. Exhibit the knowledge of error analysis in measurement. 3. Exhibit the knowledge of static and dynamic characteristics of various types of instruments. 4. Elucidate the working principle of various sensors, transducers and the concept of signal conditioning.				
Text Books 1. Measurement systems, Doebelin, E.O., Tata McGraw Hill, 2017. 2. Nakra, B.C. and Chaudhry, K.K., Instrumentation Measurement and Analysis, Tata McGraw-Hill (2009) 3rd ed				

Approved in 109th meeting of the Senate held on March 16, 2023. Revised in 112th and 114th meeting of the Senate held on March 11, 2024 and March 7, 2025 respectively.

Reference Books

1. Transducers and Instrumentation, Murthy, D.V.S., Prentice Hall of India, 2nd edition, 2008.
2. A Course in Electrical and Electronic Measurements and Instrumentation, Sawhney, A.K. and Sawhney, P., Dhanpat Rai, 18th edition, 2008.
3. Mastering Internet of Things: Design and create your own IoT applications using Raspberry Pi, Peter Waher, Packit, 2018.

Evaluation Scheme	
Evaluation Elements	Weightage %
Mid Semester Test (MST)	25-30
End Semester Examination (ESE)	40-45
Sessional (may include Assignment, Sessional (Includes Regular Lab assessment) and Quizzes Project (Including report, presentation etc.)	30